

Chapter Three

Introduction To Software Engineering

Software Engineering

- ***Software Engineering*** is composed of two words, **software** and **engineering**.
- **Software** is **more than** just a program code
 - A program is an **executable code**, having some **computational purpose**
 - Software is a collection of **executable programming code**, **associated libraries** and **documentations**
 - Software, when made for a specific requirement is called **software product**.
- **Engineering** is developing products, using **well-defined**, **scientific principles** and **methods**.

- ***Software Engineering*** is an engineering branch associated with the development of software product using well-defined scientific principles, methods and procedures with the outcome of an efficient and reliable software product.
- IEEE definition
 - The application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software
- Software engineering is a branch of computer science, which uses well-defined engineering concepts required to produce efficient, durable, scalable, in-budget and on-time software products.

Software engineering principles use two important techniques to reduce problem complexity:

- *Abstraction*: Simplifying problems by omitting irrelevant details.
- *Decomposition*: Dividing complex problems into several smaller problems and solving smaller problems one by one.

- Apply theories, methods, and tools and use them selectively
- Try to discover solutions to problems even when there are no applicable theories and methods
- Find solutions within constraints
- Software engineering is not just concerned with the technical processes of software development but also software project management and the development of tools, methods, and theories

SE Vs CS

- CS is concerned with the **theories** and **methods** that underlie computers and software systems whereas SE is concerned with the practical problems of **producing software**.
- Some knowledge of **CS is essential** for SE like some knowledge of **physics** is essential for **electrical engineers**

SE Vs System Engineering

- System engineering is concerned with all aspects of the development and evolution of systems
 - Hardware development,
 - Policy and process design and
 - System deployment, as well as software engineering
- System Engineers specify the system, define its overall architecture, and integrate the different parts to create the finished system
- But SE mainly concerned on the development of software products

Need for SE

Two reasons

1. Individuals and society rely on advanced **software systems** so as to produce **reliable** and **trustworthy** systems economically and quickly.
2. SE uses methods and techniques to make **software development cheaper**.

Presentation on

- Evolution of Software Engineering-G1
- The impact of software on economic, societal and environmental safety- G2
- Software professionalism, accreditation, certification, and licensing-G3
- Code of Ethics and professional conduct-G4
- Social, legal, historical, and professional issues and concerns-G5
- The nature role of professional societies and software engineering standards-G6